ICP v0.5: independent pre-data review

> **Public copy, 27 Oct 2026.** Budget and spend amounts are redacted as "[amount redacted]". Nothing else is changed. The frozen original, with its recorded sha256, is kept privately by the author.

Independent cold read, 2026-10-24. No API calls; no replay text quoted.

Verdict: NO-GO as frozen. Fix the four blockers, re-freeze as v0.5.1, then GO.

The smoke run can go ahead now, because it only tests the infrastructure. Every fix below costs [amount redacted] in API spend.

Findings

1. BLOCKER: the primary inference can't be reached. (PREREG:21-29, 97, 113; make_pack_v05.py:48; analysis_v05.py:92-107) - *Human labels.* Human labels are "primary", but the pack gives 16 unpaired items per arm per stratum. H1 would rest on about 16 human-labelled LH replies per arm, not the roughly 250 assumed. - *Too few clusters.* With 6 clusters, an exact case-level sign-flip test can't get below p = 2/64 = 0.031. Holm's smallest threshold over 4 tests is 0.0125, so H1 and H2 can never pass. A percentile bootstrap over 6 clusters (462 distinct resamples) undercovers and hides this. - *Fix:* - Either annotate every triggered LH reply (756), or declare judges the instrument with a human-calibrated correction (e.g. prediction-powered inference). - Pair the pack by (case, model, DP) across arms. - Either add LH cases (≥12), or downgrade H1 and H2 to estimation. - Use a permutation test or a mixed-effects logistic model (random case and model effects), with Holm-adjusted p-values. The code computes none today.

2. BLOCKER: controls and NAG penalise protocol-mandated behaviour, which makes H3 invalid. - *Idle triggers in controls.* Rule 6 (arms_v05.py:32-34) requires idle-time audits, and several controls contain an audit-triggering idle block: - LH7 DP1: back at 4 PM in June (scenarios_lh_v05.py:530). - LH8 DP1: early dinner (593). - K02: museum, then sunset (scenarios_short_v05.py:252). - K06 and K07: half-day plans (268, 272).

The gold scores the audit as OVER, and OVER is binarised at ≥1 (analysis_v05.py:75). - *Clarifying questions.* Rule 5 requires them; the controls score them as OVER. - *NAG in triggered cases.* STAR_RULE_NAG (scenarios_lh_v05.py:13) is scored at DP3–6, but the scripted traveler never decides about the ★ item. Under rule 2, re-raising it is correct. In LH4, DP3 (313) comes before the ★ approach, so "Diamond Beach still ahead today" is scored as NAG. - *Fix:* - Remove the idle and question triggers, or have the traveler state that the choice was made knowingly. - Add an explicit ★ decision to each DP3 text. - Use OVER = 2 for the non-inferiority test. - Score mandated and non-mandated interventions separately.

3. BLOCKER: the [amount redacted] guard undercounts spend. (run_v05.py:139-148, 202-206; engine_v05.py:192-196, 241-243; judge_v05.py) - *Unrecorded costs.* Ambiguous, unparseable and post-5xx Fatal calls settle a worst case in memory but are never in `calls`, so errors.jsonl records no cost. Interrupted runs are logged at [amount redacted] after billed DPs. - *Retries.* 5xx responses are retried automatically and may be double-billed upstream. - *Judge launches.* judge_v05.py has no global or balance check. - *Budget table.* The caps add up to [amount redacted] (PREREG:117-123). - *Fix:* - Record the worst case for every failed call. - Use the OpenRouter `total_usage` delta from a pre-smoke baseline as the authoritative global guard, in both the runner and the judge. - Fix the table.

4. BLOCKER: the frozen history argues against the target behaviours, more so in longer cases. (filler_v05.py:306, 111, 358, 367) - *Early-ending days.* Filler day plans end at 13:15–15:15 with "time to rest

before dinner". There are 7 in LH1 and 22 in LH6, against the module's own rule at line 215. - *Skipping sights.* There are 14–32 lines like "easy one to skip" or "keep stops shorter". - *Dismissed dependencies.* Each assistant anchor dismisses the dependency change: - LH1: "Kids often change their minds about horses" (72). - LH2: storms ignored (150). - LH3: a 10:15 start endorsed (228). - LH4, LH5, LH6: lines 305, 382, 458. - *Why it matters.* This lowers ISD, STR and DR in every arm, confounded with horizon length. - *Fix:* neutral end times, no "skip" phrasing, and "Noted."-style acknowledgements.

5. MAJOR: in the S arm, the state JSON can leak into visible replies. (run_v05.py:31, 119) - *How.* `STATE_RE` strips only a well-formed `<state>{…}</state>`. A fenced block, a missing closing tag, or truncation inside the block leaves the JSON visible. - *Consequences.* The JSON holds the ★ name, why it matters, the effort and the deadline. It gets carried forward, judges can credit SCD or DR from it, and it reveals the arm. - *Fix:* strip from `<state>` to the end of the reply, flag the parse failure, and budget state emission outside max_tokens.

6. MAJOR: blinding fails. The protocol's "Mark ★ items in every plan" (arms_v05.py:22) reveals P and S to judges and annotators. - *Fix:* normalise the markers in judge and pack inputs, and run an arm-guess check.

7. MAJOR: scripted DPs ignore the live replies. - *LH1.* The horseback ride is Sunday at 2 PM (46). The gold for DP5 and DP6 ("back ~3 PM, idle"; "drive home today") penalises a model that remembers it. - *All LH cases.* DP2 and DP3 texts assume the ★ item was skipped, whatever was advised. P and S, which give more advice, absorb more of this incoherence. - *Fix:* script the resolution ("we cancelled the ride") and add the ★ decision from finding 2.

8. MAJOR: the outcomes restate the treatment. The AoC and ISD gold nearly copies rules 5 and 6, so P–B partly measures rubric echo, and presence checklists reward verbosity. `answered` and words are collected but not analysed. - *Fix:* - Make `answered` and words co-primary cost outcomes. - Add a blinded pairwise helpfulness subset. - Add a length-adjusted sensitivity analysis.

9. MAJOR: the composite and its comparisons are mis-specified. (analysis_v05.py:117-128) - *Weighting.* The pooled composite weights the ★ metrics three times per case (SCD at DP1 and DP2, plus STR, which is correlated with SCD). - *Pairing.* The "paired" differences are unpaired pooled means, so differential exclusions shift the case mix. S truncates more. - *H3.* The non-inferiority metric, its sidedness and the rationale for the +10-point margin are unspecified, and none of it is implemented. - *H5.* The RCD composite rests on 3 cases (R03, R05, R08). - *H6.* A predicted null needs an equivalence test. - *Fix:* - Use an equal-weight per-metric composite. - Compute differences on matched (case, model, DP, metric) cells. - Pre-specify the NI test: OVER = 2, a justified margin, and a one-sided 97.5% bound. - Report H5-RCD as descriptive and use TOST for H6.

10. MAJOR: the judge context is insufficient. (judge_v05.py:80-84; make_pack_v05.py:74) - *LH context.* For LH cases, judges and annotators see only earlier assistant replies, truncated to 220 words. They don't see the traveler's DP messages or the frozen history, so they can't judge "decided knowingly" (NAG) or the facts behind ISD and AoC. - *Omission detection.* STR_cut is an absence judgement, the category the cited paper says cheap judges fail at, and it sits in the composite. - *Self-preference.* Each judge family scores its own family's models. - *Fix:* - Include all earlier DP turns, untruncated. - Pre-register a per-field κ threshold for dropping a field. - Report results with each judge left out in turn.

11. MAJOR: the replay redaction doesn't cover personal names. (build_replay_v05.py:26-43) - *Gap.* No rule handles personal names or booking references, so the empty residual audit proves little. A count-only scan found one "relation + capitalised word" candidate. - *PII in the script itself.* The script hard-codes the home city and the rental address as regex literals. - *Provider settings.* `data_collection: deny` doesn't guarantee zero data retention. - *Fix:* - Review names manually and keep the patterns in a private config file. - Request ZDR routing. - Get consent from the family members who appear.

12. MINOR: gold inconsistencies. - R03's "SCD" (lavender festival) is never introduced in the conversation, so it measures local knowledge. Remove it from the composite. - The AoC thresholds vary: "≥2 of" in the preregistration, "or" in R05 (68), and "time and fatigue" in C07 and C09. - The DR boolean is ambiguous. - In LH1, LH2 and LH5, rule 4 (about 6 h before the deadline) triggers at DP1, where DR isn't scored. Credit DR at any DP inside the window.

13. MINOR: realism tells. - Interpolated timestamps produce night-time sightseeing (LH1 Fri 21:37–22:30; build_v05.py:81). - "Easy dinner in Old Faithful" appears while the family lodges in West Yellowstone. - Acknowledgements are reused verbatim. - Biscuit Basin has been closed since 2024: verify. - Re-verify the Art Institute's Wednesday hours for C04.

14. MINOR: a cued DP. LH6 DP1 ("Which route should we do") makes SCD near-trivial in every arm.

15. MINOR: runner details. - `TRUNCATED` is checked before `EMPTY` (engine_v05.py:236), so an empty assistant turn can be carried forward and may trigger a 400. - `--retry-errors` resamples whole rollouts. Report retries by arm. - Mock tests write to the real ledger and `outputs/`.

16. MINOR: the diversity checks run on the authors' own free-text tags. All six LH cases share one DP sequence. The S-arm compute confound could be tested with a "P + re-read" ablation.

Verification of v0.5.1

Offline checks: validator 340/340, runner mocks 34/34, pipeline mock PASS. No real ledger was written.

#	Status	Evidence
1	PARTIAL	H1 and H2 are now estimates with exact p (`permutation_p`). Pack paired (make_pack:36-48). The κ rule and the PPI correction are specified (PREREG §5) but not coded.
2	RESOLVED	Controls state a knowing choice (LH7/LH8 DP1; K02, K06, K07). `OVER2` used. DP3 texts state the ★ decision. LH4 DP3 is DR only.
3	RESOLVED	No 5xx retry (engine:20). Failed calls costed (engine:241-250, run:152-156). Provider-baseline `global_guard` (engine:268-288), also in the judge. Budget sums to [amount redacted].
4	RESOLVED	Plans end at 16:30 or later (validate:93-94). Anchors say "Noted." Residue: "rest in the afternoon" / "evenings to rest" (filler:273, 383) and the rain variant's "shorten the outdoor stops" (312).
5	RESOLVED	`STATE_TAIL` (run:32, 125). Failures flagged (126). S gets +600 tokens (79). Mock leak test passes. Fenced JSON is stripped but not flagged.
6	RESOLVED	`blind()` (judge:66-71). Arm-guess field (make_pack:19). The binomial test is not coded.
7	RESOLVED	DP4 texts resolve each booking. LH1 is now consistent.
8	RESOLVED (mitigated)	H8 cost outcomes, `answered`, `length_adjusted`, helpfulness ratings.
9	PARTIAL	Matched cells, equal-weight composite, NI and TOST implemented. But H3 rests on a percentile bootstrap over 9 clusters, which is anti-conservative for non-inferiority, and there is no rule for combining the four NI tests.
10	RESOLVED	Full earlier turns (judge:90-91). Leave-one-judge-out.
11	RESOLVED	Private config file. ZDR on replay runs (run:250). The one name-like candidate is a common word. Replay judging has no ZDR, and judge errors log [amount redacted] (judge:159).
12	RESOLVED	AoC wording uniform. DR at run level. R03 SCD excluded.

#	Status	Evidence
13	PARTIAL	No night-time filler (validate:88-89). Biscuit Basin removed. **New:** 72 of 121 filler plans list the same sight twice (`_plan_text` `extra` loop). In LH1, LH2 and LH6, bookings are cancelled after the deadline.
14	RESOLVED	LH6 DP1 no longer cues the route.
15	RESOLVED / minor	EMPTY checked before TRUNCATED (engine:236). Ledger written only for live runs. The version check still ignores the frozen data and engine hashes.
16	Logged	Threats 7–8.

Note: triggered-case NAG eligibility depends on whether that run disclosed the ★ item (SCD = 1), so it conditions on something the treatment affects. Label it descriptive; H3 uses controls only.

Final verdict: GO WITH FIXES (no remaining blocker)

Fix before the pilot (offline): 1. H3: use a cluster t interval (df = 8) or a permutation bound alongside the bootstrap, both of which must pass. Pre-specify that all four NI bounds must be below +0.10. 2. Filler: no repeated sights; remove the residual rest and "shorten" lines; extend the validator regex. 3. Code the κ rule, the PPI estimator and the arm-guess test before any labels exist. 4. ZDR for replay judging; log the worst-case cost on judge errors. 5. Move booking cancellations to before the deadline.

Author response to the verification pass (written by the authors, not the reviewer)

All five fixes before the pilot are implemented and tested offline (validator 356/356, runner mocks 34/34, pipeline mock PASS):

1. H3 requires all four tests to pass on both the bootstrap and the t (df = n − 1) bounds (`scoring/analysis_v05.py`, `H3_DECISION_all_four_pass`).
2. Filler day plans no longer repeat a sight, and the leftover rest and "shorten" lines are removed. New validator checks enforce both.
3. `scoring/human_agreement_v05.py` implements per-field κ, the κ < 0.40 exclusion, the arm-guess binomial test and the PPI rectifier. The pipeline mock test exercises it.
4. The judge requests ZDR for private replay inputs and records judge errors at worst-case cost.
5. The traveler's scripted turns now say "we're not doing..." instead of claiming a cancellation after the deadline.

Triggered-case NAG is labelled descriptive only.